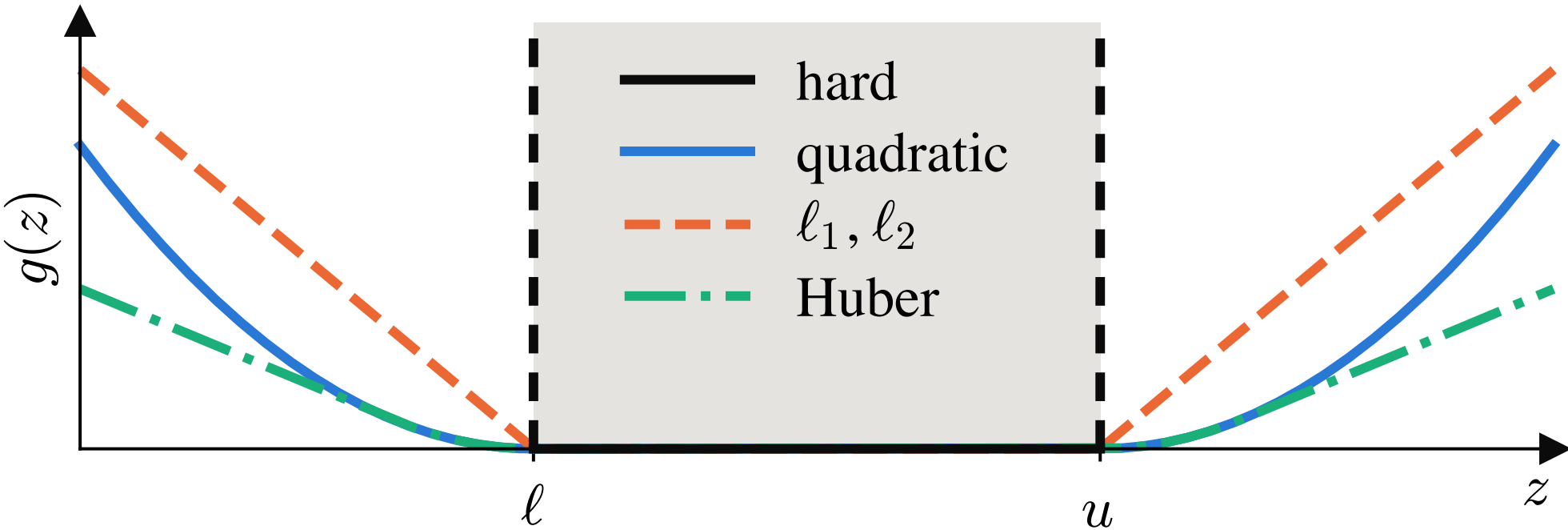
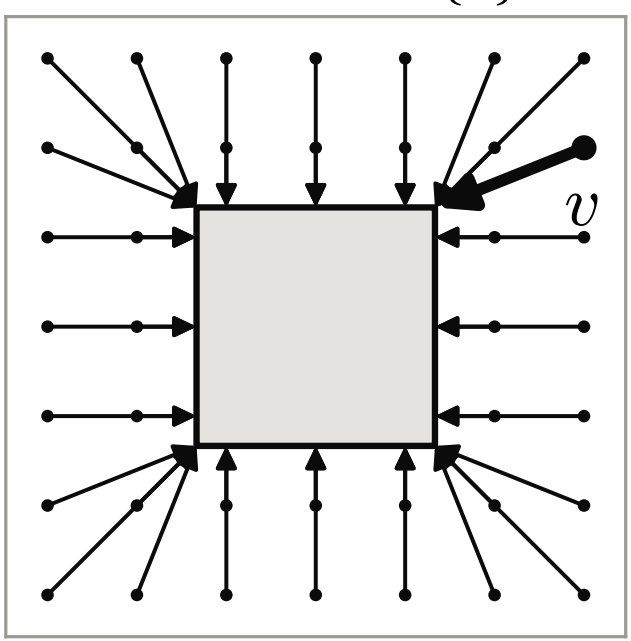
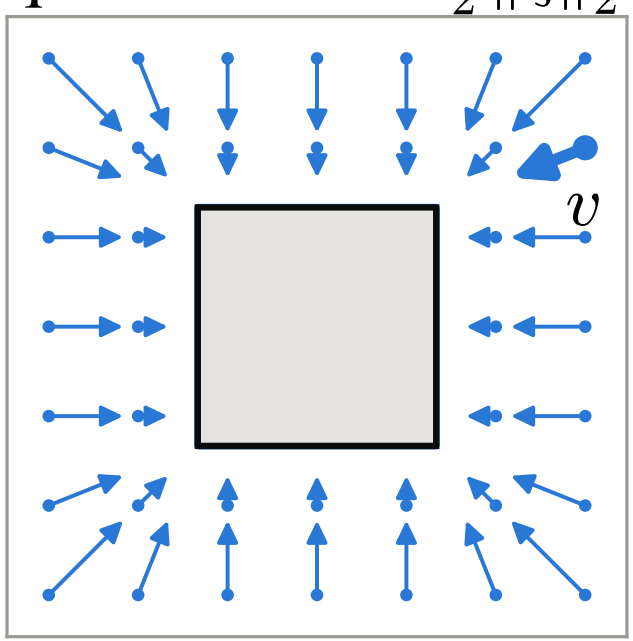
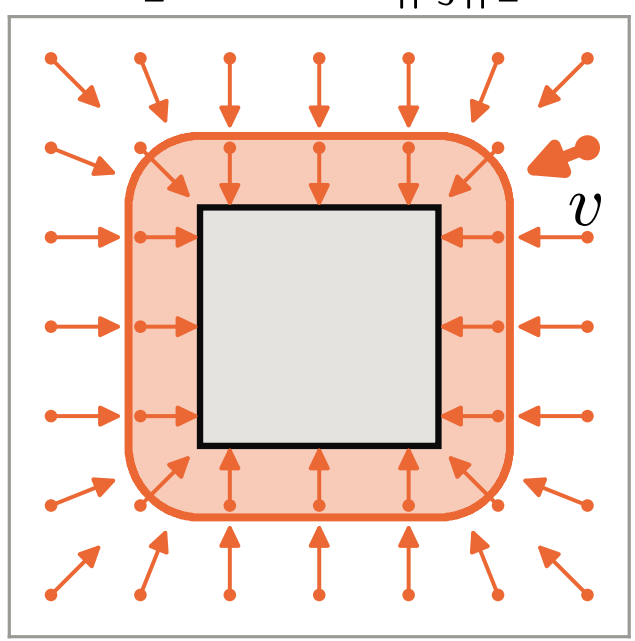


hard: $\Phi = I_{\{0\}}$



quadratic: $\Phi = \frac{\alpha}{2}\|\xi\|_2^2$



ℓ_2: $\Phi = \alpha\|\xi\|_2$